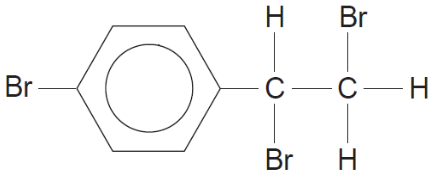
Question				**Marking details**	**Marks available**					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
10	(a)			**Indicative content** mass of carbon in 5.35 g of compound **B** is 1.50 g molecular formula is $C_8H_7Br_3$ M_r is 343 there is a chiral carbon atom formula of silver bromide is AgBr mass of bromine in 4.77 g of silver bromide is 2.03 g NaOH hydrolyses aliphatic C—Br bonds % Br in hydrolysis product is 46.6 % Br in compound **B** is $\frac{3 \times 79.9 \times 100}{343}$ = 69.9 therefore two of the three bromine atoms in compound **B** are removed ⇨ they must be in the side chain bromine atom not removed must be on the ring compound **B** is  accept aromatic bromine in any position		3	3	6	2	2

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
				5-6 marks All the information used effectively; sufficient explanation to deduce the correct structure *The candidate constructs an articulate, integrated account, which shows sequential reasoning. The answer fully addresses the question with no irrelevant inclusions or significant omissions. The candidate uses scientific conventions and vocabulary appropriately and accurately*. **3-4 marks** Most of the information used effectively; some features of the structure correctly identified and explained *The candidate constructs an account correctly linking some relevant points showing some reasoning. The answer addresses the question with some omissions. The candidate usually uses scientific conventions and vocabulary appropriately and accurately*. **1-2 marks** Some of the information used correctly; simple conclusions drawn *The candidate makes some relevant points showing limited reasoning. The answer addresses the question with significant omissions. The candidate makes limited use of scientific conventions and vocabulary.* **0 marks** *The candidate does not make any attempt or give a relevant answer worthy of credit.*						

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		eight —OH groups react 8 × 0.0700 mol of ethanoic anhydride needed = 0.560 mol (1) mass = 102 × 0.560 = 57.1 g minimum volume = $\frac{57.1}{1.08}$ = 52.9 cm^3 (1)	1	1		2	1	
		(ii)		award (1) each for any two of following percentage yield reaction rate / time needed need of a catalyst energy costs – must be qualified (and not related to heating)			2	2		
	(c)	(i)		red-brown precipitate/solid	1			1		1
		(ii)		aldehyde / CHO	1			1		1
				Question 10 total	**3**	**4**	**5**	**12**	**3**	**4**